NMEO-OP | Oilseed Value Chain Platform

Complete Setup, Dependencies & Execution Guide (React + JavaScript)

1. System Prerequisites

• Node.js: v18.0.0 or v20.x+ LTS (https://nodejs.org)
• npm: v9.x or v10.x+ (bundled with Node.js)
• Git: Any recent version (https://git-scm.com)
• Web Browser: Google Chrome, Microsoft Edge, Brave, or Firefox

2. React + JavaScript Dependencies (package.json)

• **react (^19.2.8):** Core React 19 UI component framework
• **react-dom (^19.2.8):** React DOM rendering package for web browsers
• **react-router-dom (^7.18.3):** Client-side routing and multi-page navigation
• **@supabase/supabase-js (^2.49.1):** Supabase client for Auth, PostgreSQL DB, and file storage
• **lucide-react (^0.475.0):** Modern icons for agricultural & value-chain interfaces
• **recharts (^3.10.1):** Interactive charts for AI price/demand forecasting
• **leaflet (^1.9.4) & react-leaflet:** OpenStreetMap engine for regional crop health maps
• **axios (^1.20.0):** HTTP client for ML microservices & REST APIs
• **vite (^8.2.2):** Ultra-fast build tool and development server (devDependency)
• **@vitejs/plugin-react (^6.1.1):** Official Vite React JSX compiler & fast refresh plugin

3. Step-by-Step Installation & Run Commands

Step 1: Clone Repository

```
git clone https://github.com/your-repo/Oilseed-valueChain.git
```

Step 2: Enter Project Directory

```
cd Oilseed-valueChain
```

Step 3: Install All Dependencies

```
npm install
```

Step 4: Start Development Server

```
npm run dev
```

Step 5: Build for Production

```
npm run build
```

4. Environment Variables Configuration (.env.local)

Create a file named .env.local in the root folder with:

```
VITE_SUPABASE_URL=https://aiobuyiuarozsbfzonam.supabase.co
VITE_SUPABASE_ANON_KEY=your-supabase-anon-key-here
VITE_STORAGE_CROP_IMAGES_BUCKET=crop-images
VITE_STORAGE_DOCUMENTS_BUCKET=documents
VITE_STORAGE_ML_MODELS_BUCKET=ml-models
```

5. Database & Auth Setup in Supabase

1. Open Supabase Dashboard -> SQL Editor -> New Query.
2. Run supabase/migrations/00000_complete_week1_schema.sql to create all 14 tables.
3. Run supabase/auth_trigger.sql to configure controlled Sign Up vs Sign In RLS.
4. In Supabase Authentication -> SMTP Settings, configure Gmail SMTP (smtp.gmail.com : 587).